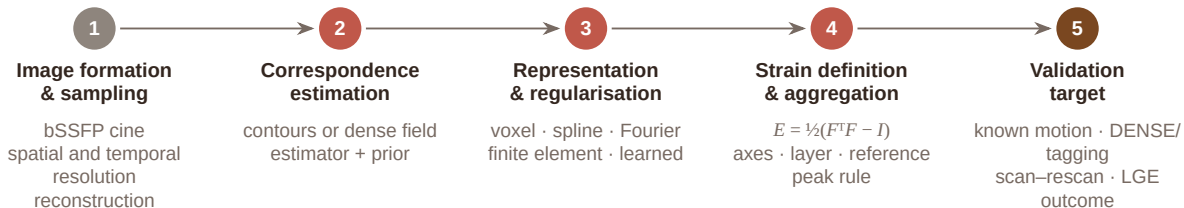


## a From cine images to a regional strain report

● observed in cine ● inferred ● reference



## b Three failure points, each testable on its own

### ① Input sufficiency

Do the images  
contain the cue?  
Test: re-image the  
same motion more  
finely or with more  
views

### ② Estimation

Does the optimiser  
or network recover  
it?  
Test: vary  $\alpha$  (penalty  
weight) with images  
fixed

### ③ Representation capacity

Can the basis  
express the lesion?  
Test: project the  
known field onto the  
function space

### If a known focal deficit is under-recovered

recovers when  $\alpha$  is reduced → **estimation**

still fails after projecting the  
known field → **representation**

recovers only after re-  
imaging → **input information**

## c

*Validation must follow the finest claimed spatial and temporal scale, not inherit credibility from an upstream task.*

→ output feeds the next step

① failure point at step n

Conceptual schematic; no study data.